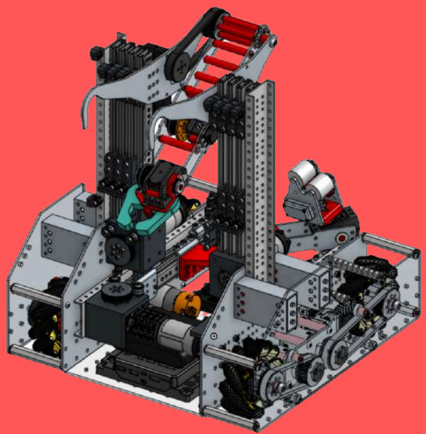
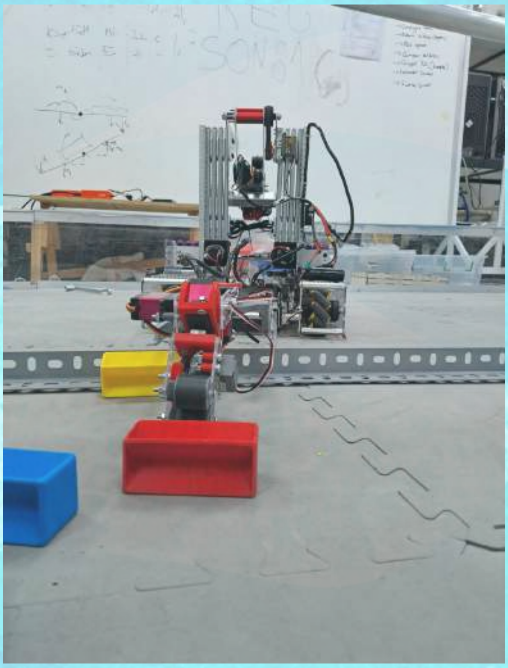
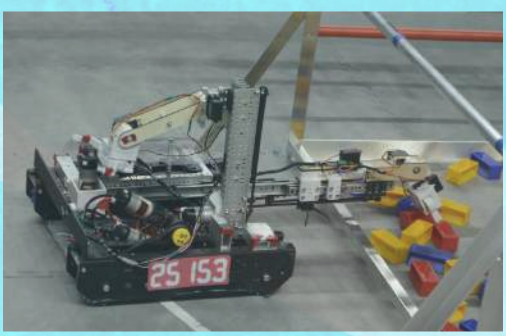
Cartesian #25153

"I think, therefore I can."







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METU Foundation Developement Schools

FIRST TECH CHALLENGE

FİKRET YÜKSEL FOUNDATION

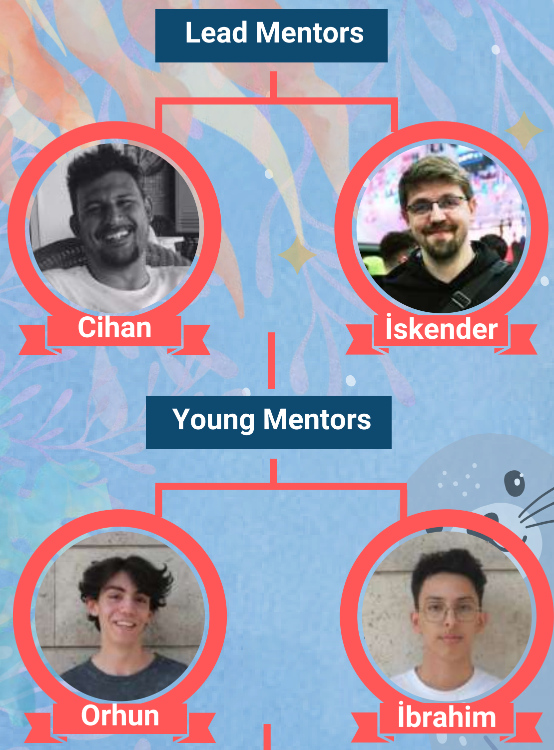
INTO THE DEEP

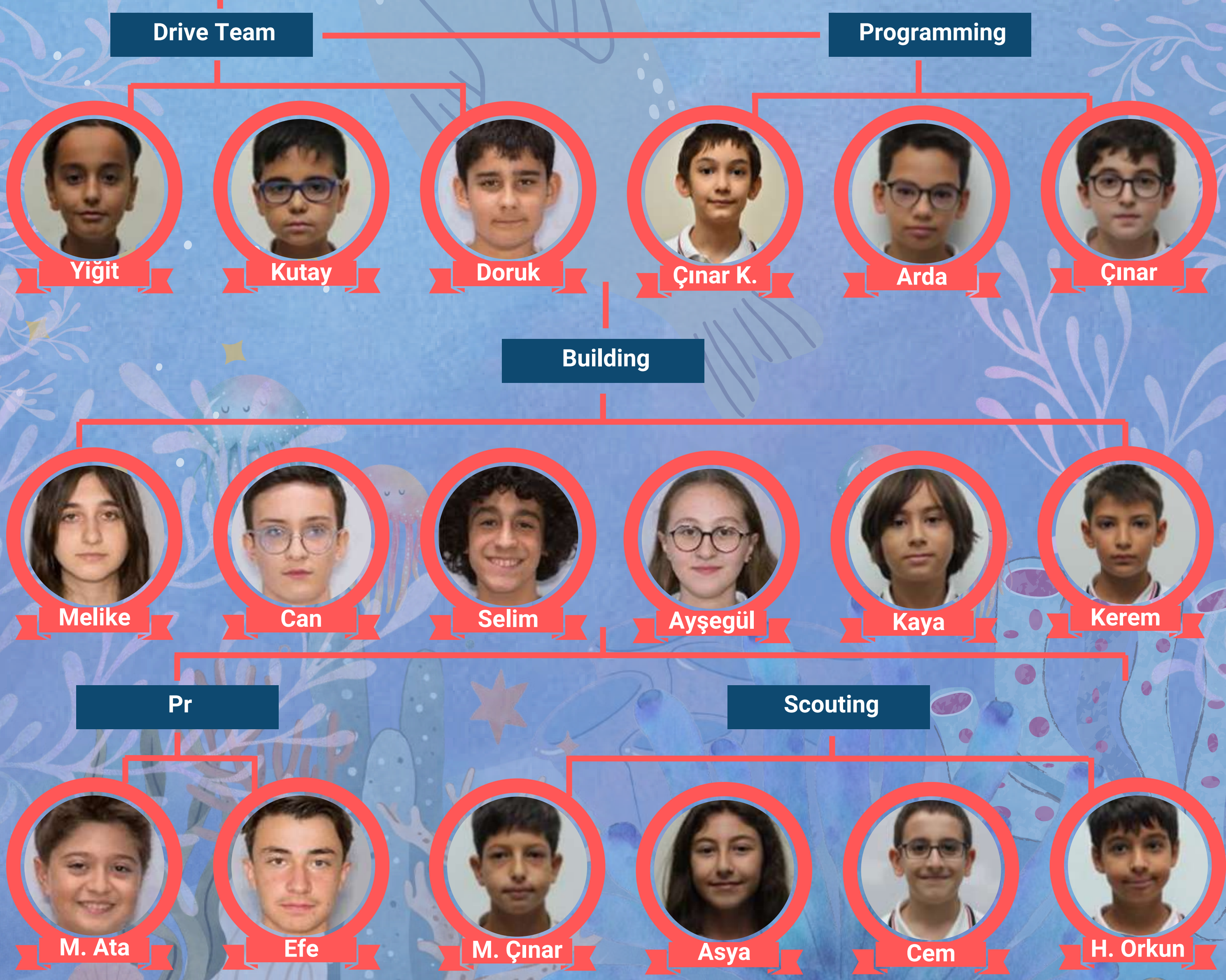
WHO WE ARE?

I think, therefore I can.

We are Cartesian Robotics, a dynamic and passionate team composed of 7th and 8th grade students from METU Development Foundation Schools, from Turkey. Supported by our young mentors, teachers, and families, we operate as a collaborative and multi-faceted community where learning, creativity, and innovation come together. Our mission is to empower middle school students through hands-on STEM experiences while instilling the core values of FIRST, including the principle of Gracious Professionalism—competing fiercely, yet always with kindness, respect, and empathy.

We believe that robotics is not just about building machines, but about building character, solving real-world problems, and growing together as a team. Through our outreach efforts, technical development, and teamwork, we aim to spark curiosity and inspire the next generation of changemakers. At the heart of everything we do lies our motto: "I think, therefore I can."





TEAM STRUCTURE

Cartesian Robotics operates as a well-organized team composed of 7th and 8th grade students, guided by a dedicated network of young mentors, teachers, and supportive families. Our team is structured into several key departments, each playing a crucial role in the success of our season.

Building Team is responsible for designing, prototyping, and building the robot. Using tools such as CAD, 3D printers they transform ideas into functional systems.

Programming Team develops all robot code, including autonomous routines, driver control systems, and sensor integration. They ensure the robot is both smart and responsive on the field.

Public Relations (PR) Team manages social media, sponsorships, and outreach. They also handle branding, visual design, and organize events that connect us with our community.

Scouting Team analyzes game rules, develops match strategies, and supports decision-making during competition. They also collaborate with other teams to adapt and improve.

MISSION, VISION & GOALS

At Cartesian Robotics, our vision is to empower young minds to become confident problem-solvers and future innovators by combining theoretical knowledge with real-world applications. Our mission is to create a learning environment where students can freely explore STEM, develop creativity, and grow as responsible individuals—embracing equality, collaboration, and the spirit of FIRST.

We aim to:

- Build technically advanced and creative robots each season,
- Foster a love of STEM in our community through outreach,
- Grow as a team that supports each other with kindness and respect,
- Encourage leadership, curiosity, and resilience in all members.

We proudly live by our motto: **I think, therefore I can.**

WHAT MAKES US UNIQUE IN FTC

What sets Cartesian Robotics apart is our student-driven spirit, paired with a mentorship model that empowers middle school students to take the lead in meaningful engineering projects. From locally manufactured robot parts to in-house developed prototypes and custom-coded control systems, we constantly seek original and sustainable solutions that reflect both our creativity and our commitment to progress.

Our collaborative culture includes not only students but also families, alumni, and international connections. We actively mentor other teams, contribute to open-source platforms, and organize awareness campaigns and outreach events to spark interest in STEM. Our dedication to making a broader impact turns every challenge into a shared opportunity for growth.

This year, our robot Hilda, named in honor of a Mediterranean monk seal we adopted, represents both the technical excellence and environmental consciousness at the heart of our team. Inspired by the “Into the Deep” theme, Hilda reflects our belief that engineering should not only solve problems—but also serve a purpose.

SUSTAINABILITY

At Cartesian Robotics, sustainability begins with investing in human potential. Our team consists of middle school students, selected from within our school for their curiosity, dedication, and enthusiasm for STEM. Through FTC, we equip them with essential skills in engineering, programming, and collaboration—building a strong foundation for their transition into our FRC high school team. This process is guided by our young mentors and lead mentors, ensuring that knowledge and leadership are passed down through each generation. This layered mentorship structure strengthens continuity and empowers students to grow into confident leaders.

On the financial side, our team is supported by our school’s resources, along with generous contributions from parents who assist through donations and volunteering. To promote financial sustainability and independence, our PR team actively works to secure sponsorships and partnerships throughout the season. These combined efforts ensure that we can continue to innovate, grow, and compete while also teaching students the real-world importance of planning, communication, and financial responsibility.

PARTNERSHIPS & SPONSORS

At Cartesian Robotics, we are proud to be supported by a strong network of sponsors and partners who believe in the power of youth-driven innovation. Our main sponsor, METU Development Foundation School, plays a vital role in sustaining our team by providing institutional support, facilities, and encouragement throughout the season. In addition, our team members' families contribute in meaningful ways—through donations, logistics, and active involvement in team activities.

Beyond this, we have also gained the support of several private companies who contribute to our mission through sponsorship. These sponsors typically assist in two key ways: by supplying materials and equipment necessary for building and developing our robot, or by providing financial sponsorship to support our competition expenses. Our Public Relations (PR) team shares our sponsorship package file with potential partners, allowing them to select the level and type of support they wish to offer. Thanks to these valuable collaborations, we are able to push the limits of our creativity and engineering, while also building meaningful relationships with the professional world.

METU Foundation Developement Schools
NFRProducts
SIEMENS Solid Edge
BOEING
CREALITY
Beta
IKOM
mikropor
edata
SONUÇ YAYINLARI
optima
YAMANTÜRK VAKFI
ATADA

OUR JOURNEY WITH GROWTH AND SUCCESS

Since our rookie year in 2023, Cartesian Robotics has rapidly grown from a passionate group of students into an award-winning FTC team. With each season, we've challenged ourselves to think deeper, build smarter, and collaborate stronger. Our journey is marked by constant improvement, international experience, and national recognition.

Early Achievements

Before official FTC events were launched in Türkiye, we actively took part in off-season regionals and international competitions to gain experience and test our potential. At the FTC Thessaloniki Regional, we earned 3rd Place, along with the Innovate Award (3rd Place) and the Design Award (2nd Place)—our first international recognition. Domestically, we placed 2nd in the FTC Istanbul Off-Season, and again in FTC Istanbul Season 2, where we also received the Design Award (2nd Place) and the Innovate Award (1st Place). These early milestones formed the foundation of our competitive identity and gave us the confidence and skills to succeed in official FTC seasons.

This Season's Regionals

This season, with Türkiye's official FTC structure in place, we stepped into a more competitive landscape—and rose to the challenge. At FTC Istanbul Üsküdar Regional 1, we were honored with the Control Award, recognizing the strength of our programming, sensor integration, and precise automation systems.
Shortly after, we competed in FTC Istanbul Üsküdar Regional 2, where we earned the Inspire Award (2nd Place)—one of the most prestigious recognitions in FTC. Being selected as part of the Finalist Alliance also demonstrated our ability to collaborate effectively with other teams and contribute strategically to shared success.

National Victory

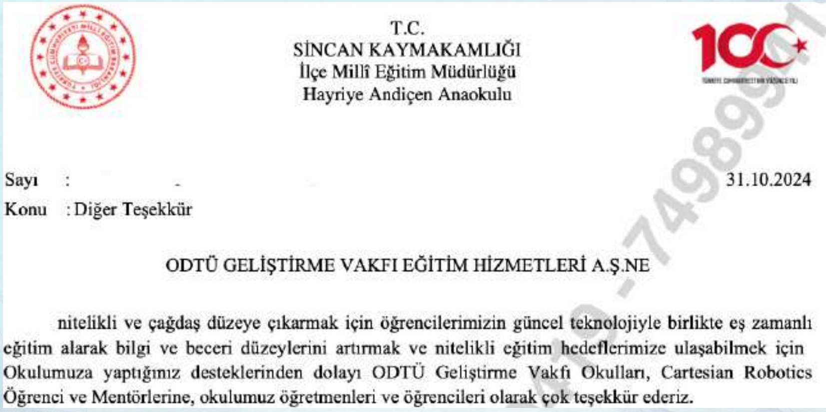
Our journey reached its peak at the FTC Türkiye Championship, where we were named the Winning Alliance Captain—a title that celebrates our leadership and performance at the national level. Additionally, we proudly received the Think Award, highlighting the clarity and depth of our engineering process and documentation.
These achievements reflect not only our technical growth but also the solid foundation we've built through mentorship, discipline, and learning from every challenge.

In line with our mission to empower young minds through STEM, our vision of spreading innovation, and our goal of creating meaningful impact both inside and outside the field, we believe robotics is more than competition—it's a bridge between education and real-world change. Guided by these principles, we proudly take initiative in organizing events that spark curiosity, promote collaboration, and give back to our community.

Empowering the Future – Sincan STEAM Project #1

In line with our commitment to equal access in STEM, we identified a local school in Sincan, Ankara, that faced technological limitations. Our team donated 5 laptops and 30 PCs to this school and took the initiative to establish a STEAM workshop within its facilities. This space now allows students from kindergarten to middle school to engage with hands-on science and engineering activities. With this effort, we aimed not only to bridge the digital divide but to ignite lifelong curiosity and opportunity in communities that need it most.

An official letter of appreciation was presented to Cartesian Robotics by the Sincan District Directorate of National Education:



T.C.
SİNCAN KAYMAKAMLIĞI
İlçe Millî Eğitim Müdürlüğü
Hayriye Andiçen Anaokulu

Sayı :
31.10.2024
Konu : Diğer Teşekkür

ODTÜ GELİŞTİRME VAKFI EĞİTİM HİZMETLERİ A.Ş.NE

nitelikli ve çağdaş düzeye çıkarmak için öğrencilerimizin güncel teknolojiyle birlikte eş zamanlı eğitim alarak bilgi ve beceri düzeylerini artırmak ve nitelikli eğitim hedeflerimize ulaşabilmek için Okulumuza yaptığınız desteklerinden dolayı ODTÜ Geliştirme Vakfı Okulları, Cartesian Robotics Öğrenci ve Mentörlerine, okulumuz öğretmenleri ve öğrencileri olarak çok teşekkür ederiz.

Smart Solutions Through Engineering #2

This year, when our school planned to procure external tablet storage carts for newly purchased tablets, Cartesian Robotics proposed a more sustainable and cost-effective alternative that would also benefit future school projects. We suggested investing in a laser cutting machine for our workshop—a proposal that was accepted and successfully implemented. Using this new tool, our team designed and manufactured custom tablet cases in-house, demonstrating not only technical skill but also our commitment to sustainability and innovation.

This initiative saved resources, encouraged local production, and provided a practical example of how student-led engineering can address real-world needs. More than just a technical accomplishment, the project became a symbol of how the right tools, creativity, and a solution-focused mindset can empower students to create meaningful, lasting impact within their community.

Face-to-Face Children's Workshops #3

At the ATO Congresium cultural fair, we hosted a hands-on robotics experience for hundreds of children, families, and educators. Through interactive stations, visitors had the chance to build and drive robots, ask questions, and explore the world of engineering in a joyful, engaging environment.

This experience showed us that robotics can connect generations, spark creativity, and open doors for those who had never imagined themselves in STEM.

STEM in the Classroom – Inspiring 4th Graders #4

To introduce STEM concepts at an early age, our team organized an in-class event with 4th-grade students at a local primary school. We prepared age-appropriate science and engineering challenges, encouraging students to explore problem-solving through interactive, hands-on activities. The excitement on their faces reminded us why outreach matters—and how one small spark can light a lifelong passion for discovery.

OUTREACH & COMMUNITY IMPACT

Science Fair – Bringing Robotics to Life #5

At our school's annual Science Festival, we transformed our booth into an interactive space where the excitement of robotics and the spirit of FIRST came to life. Students from elementary to high school, as well as parents, educators, and school administrators, were invited to engage directly with our robot—controlling its movements, exploring its subsystems, and discovering the engineering process behind it. Our team members actively explained the design journey, shared insights into competition strategy, and highlighted the values that guide our work both on and off the field. More than just a showcase of technology, this event was a celebration of collaboration, curiosity, and innovation—core principles of the FIRST ethos. By opening our workspace to the wider community, we inspired others to see robotics not just as machines, but as powerful tools for learning, leadership, and meaningful impact. It was a moment where students became mentors, and visitors became future innovators.

600+

Hours Dedicated.

3000+

Students and Parents Impacted.

"We believe every connection we make and every idea we build has the power to inspire change—not just in technology, but in lives."

ScienceX – Showcasing Purpose Through Progress #6

During the ScienceX event, our team took the stage to deliver a public presentation that highlighted the season's key milestones, including our technical innovations, outreach activities, and team development journey. It was not just a showcase of achievements, but a moment of reflection on our identity, mission, and the impact we aim to create. We emphasized how each robot part represented teamwork, each outreach event embodied community, and each challenge we overcame taught us resilience. Through personal stories, visuals, and technical insights, we connected with the audience on both an emotional and intellectual level. The session allowed attendees to grasp the broader vision of FIRST, and how robotics can be a platform for growth, leadership, and purpose-driven innovation.

Future Plans

We aim not only to build stronger robots but to build stronger connections—with communities, schools, and future innovators. Here's what we're planning next:

- We plan to organize mobile STEM workshops for underserved schools in rural areas.
- We aim to develop a free online platform with STEM tutorials and outreach resources.
- We intend to design and distribute beginner-friendly robotics kits to promote hands-on learning in local communities.

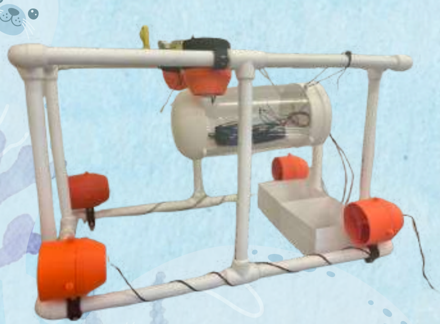
"We build today with tomorrow in mind."

This year's FIRST Tech Challenge theme, "Into the Deep," encouraged teams to dive beneath the surface—both literally and figuratively. Inspired by this theme, Cartesian Robotics carried out a series of impactful activities that combined engineering, environmental awareness, and community outreach. From designing an underwater drone to assist in the search for endangered Mediterranean monk seals, to organizing awareness campaigns and educational sessions, we aimed to connect innovation with meaningful action. These initiatives reflect our commitment to making a difference—both on the competition field and in the real world.

Hope in the Deep #1

As Cartesian Robotics, we organized a meaningful charity fair to raise awareness about the endangered Mediterranean monk seals and support their conservation through the Underwater Research Society (SAD). Our team combined innovation, teamwork, and community engagement to make this event impactful. Every product sold was carefully designed and prepared by our club members, reflecting their creativity and dedication. Throughout the fair, our PR team actively promoted the cause by distributing badges and informative materials, increasing visibility and awareness.

The fair not only raised awareness but also showcased our ability to unite for a meaningful cause through education and outreach. As a result, we successfully adopted two Mediterranean monk seals, "Kazım Sezgin" and "Hilda"—one on behalf of our club and another representing all teams at the FTC Turkey Championship. This symbolic act connected our technical mission with a real-world environmental cause, emphasizing that robotics can serve both innovation and impact. The adoption also inspired many visitors and participants to learn more about marine life and take action in their own communities. Through this initiative, we proved that even a student-led team can make a difference when guided by purpose, compassion, and collaboration



Mission: Monk Seals #2

As part of the "Into the Deep" theme, we initiated a project to design an underwater drone aimed at searching for Mediterranean monk seals. Our first step was reaching out to the Middle East Technical University Institute of Marine Sciences to gather essential information. Using the insights we received, we developed multiple design concepts and evaluated different models. After careful analysis, we selected the most effective design and successfully completed its prototype.

From Awareness to Impact #3

To increase awareness about Mediterranean monk seals, Cartesian Robotics collaborated with the Middle East Technical University Underwater Society, inviting them to our school for an educational session. The presentation explored Turkey's marine biodiversity, threats to species, and conservation efforts, with a special focus on protecting monk seals. Over 600 students attended, engaging in a Q&A session that deepened their understanding of marine conservation. This initiative not only raised awareness but also inspired students to take an active role in protecting the environment, leaving a lasting impact on our community.

At Cartesian Robotics, we strive to promote a culture of collaboration, inclusivity, and growth. Through outreach and team-to-team support, we bring the principles of Gracious Professionalism to life—competing with excellence while treating others with respect, empathy, and a willingness to help.
From mentoring rookie FTC teams and supporting schools with limited access to resources, to collaborating with world champion teams and creating knowledge-sharing platforms, we are committed to advancing the values of the FIRST community. Each partnership, visit, and conversation helps us grow not only as engineers, but as responsible leaders and lifelong learners.

As part of our commitment to fostering a more connected and collaborative FTC community, we created a dedicated forum on the *NFR Forums* platform. Through this space, we regularly share our season progress, technical insights, outreach activities, and design solutions—allowing other teams to learn from our journey while also giving us a chance to reflect on our development.

25167 – Technorams, İstanbul

Shared expertise and guidance on robotics design and development.

215564 – Mid-Chaotics, Ankara

Supplied field materials and hosted them in our workshop for hands-on support and technical guidance.

25156 – Robest, Ankara

Hosted them in our workshop for additional mentoring and technical assistance.

25155 – Ocean Warriors, İzmir

Provided mechanical parts such as gears and servo programmers.

24501 – IdignaFTC, Diyarbakır

Offered material support to help them progress in competitions.

25601 - Pioneers, Mersin

From the start, we supported the formation of the FTC team at Mersin by guiding them through team structure and roles. We donated field equipment and robot materials, and provided introductory training in mechanics and programming. This season, they successfully joined the FTC Istanbul Regional as a rookie team, proudly taking their first step into the FTC community.

This season, we held several international meetings that expanded our technical knowledge, strategic thinking, and global awareness. We met with **Team 23863 - Electrohephathenix** from India to discuss FTC preparation strategies, public relations techniques, and ways to strengthen community engagement.
We also joined an online session with two world-renowned team from the United States and Romania: **Team 11212 – The Clueless** and **Team 19066 – AiCitizens**, the reigning world champions. They generously shared strategies on autonomous optimization, match planning, and team organization, helping us refine our internal processes and elevate our performance.
In addition, we connected with T**eam 14380 – Blue Botbuilders** from Australia, where we explored each other's robot design philosophies, game analysis approaches, and community impact efforts.

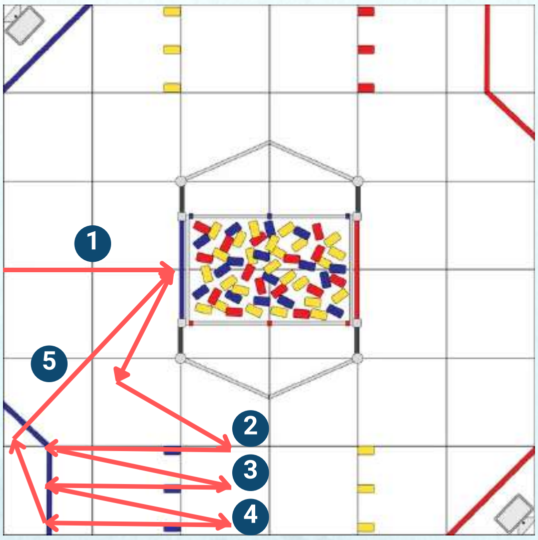
GAME STRATEGY

ALLIANCE & SCOUTING STRATEGY

At Cartesian Robotics, alliance synergy is key to our match success. We carefully scout our potential partners, prioritizing teams with strong high basket capabilities, allowing us to focus on hanging specimens in the high chamber. However, if we observe our alliance cannot consistently play the basket, our robot is designed to be strategically adaptable. We can switch to a basket-focused autonomous and TeleOp strategy, showcasing our robot's flexibility and commitment to alliance success. Every decision is backed by detailed scouting, match data analysis, and driver adaptability.

To move in better synchronization with our partners during the autonomous period, we developed a flexible and easily adjustable coding system. This allows us to tailor autonomous strategies specifically to our partners, particularly during final matches.

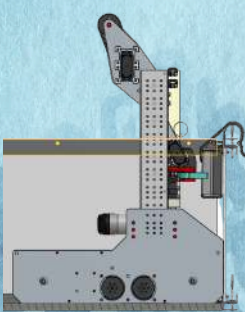
Autonomous Strategy

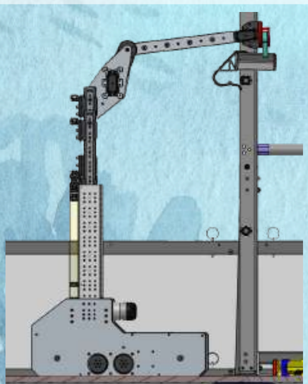


In the autonomous period, our robot prioritizes fast and efficient specimen hanging in the high chamber, with a target of 5 pieces hung before TeleOp begins. This strategy maximizes early-game scoring and allows for strong field positioning. Using our modular and partner-adaptive code structure, we can quickly switch between hanging and basket autos, depending on our alliance's capability. This adaptability has proven essential in playoff matches, especially during the FTC Turkey Championship, where synchronization with alliance partners created early scoring momentum.

End-Game Strategy

For the end game, we've engineered our robot to perform a high hang in under 3 seconds, offering valuable bonus points when needed. While our regional strategy focused on maximizing point gain through last-minute sample hangs, our championship-level approach prioritizes a high-speed, secure end-game hang, ensuring maximum point efficiency in critical matches.

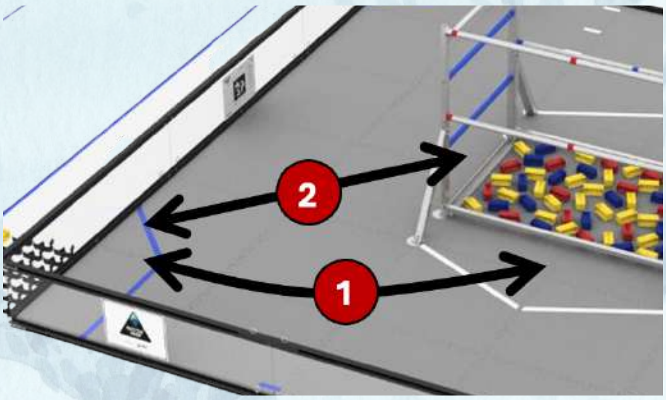




GAME STRATEGY

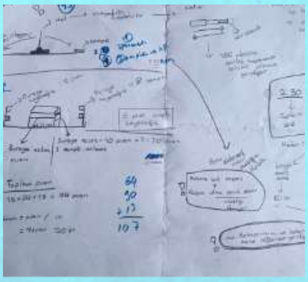
TeleOp Strategy

In TeleOp, we adopt a submarine-focused collection strategy, aiming to gather as many specimens as possible and hang them efficiently. If specimen availability is low or alliance coordination shifts, we are equipped to transition to high basket scoring, using the yellow samples either from the human player or directly from the submersible. Our robot's reliable and quick intake mechanism allows us to maintain flow and maximize scoring through flexible adaptation. Driver communication and real-time strategy shifts are central to our in-game success.



ENGINEERING PROCESS

At Cartesian Robotics, we believe great engineering comes not from getting everything right the first time, but from learning, improving, and iterating. With our mindset of "learn from mistakes and develop," we view every challenge as a chance to grow. Our process is driven by creative problem-solving, testing, and continuous refinement, where each failure brings us closer to success.



1 Brainstorming

Our process begins with open team brainstorming, where all members—from mechanics to programmers—contribute ideas and possible solutions based on the game challenge. We evaluate game elements, scoring potential, and driver practicality. Multiple solutions are sketched out before narrowing down to the most promising ones.

2 Design & Plan

After identifying a core concept, our mechanical and programming teams collaborate to create CAD models, flowcharts, and test plans—carefully weighing trade-offs like strength vs. weight or complexity vs. reliability. Using Onshape as our cloud-based CAD platform, we were able to collaborate in real time, explore multiple design paths, and prioritize solutions that allowed for easy iteration, knowing that our first version would rarely be our last.

3 Build Prototype

We rapidly build functional prototypes using 3D printing and laser cutting, focusing on testing mechanism logic, movement range, and interaction with game elements. The addition of laser cutting significantly accelerated our process, enabling fast iterations and allowing us to explore more design ideas in less time. At this stage, our goal isn't perfection—it's to learn what works and refine what doesn't.

4 Test & Analyze

Each prototype was tested for durability, consistency, and simplicity, using side-by-side comparisons and controlled strength tests. For example, we printed identical parts in PLA and PETG and measured their performance under stress to determine the best material. These tests helped us identify key improvements and eliminate inefficiencies early.

5 Iterate & Improve

Based on test results, we returned to Onshape and reworked designs—adjusting dimensions, changing materials, or improving mechanical fit. Iteration became the most important stage of our process, allowing us to refine each component step by step. Subsystems like our In-Drop and Intake went through several cycles before reaching consistent performance.

Finalize & Fine Tune

Once a design consistently passed testing and integrated smoothly with other subsystems, we finalized its construction with durable materials. Clean assembly and system harmony became priorities at this stage.

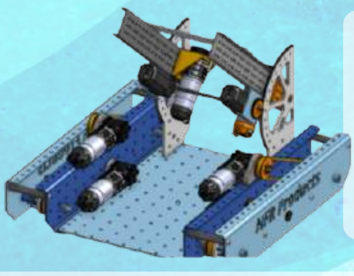
We dedicated much of our time to fine-tuning each subsystem, carefully optimizing every detail to ensure peak performance.

6

We documented each prototype version, design change, and outcome, helping us improve internally and prepare for judging. Reflection turned every mistake into a valuable insight—the heart of our engineering.

BUILT THROUGH ITERATION

At Cartesian Robotics, iteration is more than a method—it's our mindset. We believe that every prototype, test, and failure is a step toward a better solution. Instead of chasing perfection on the first try, we focus on learning fast and improving faster, continuously refining our designs through cycles of testing, feedback, and innovation. This approach has allowed us to transform initial ideas into high-performing systems, all while building resilience, adaptability, and deeper technical understanding within our team.

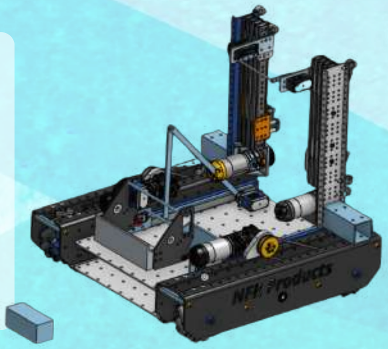


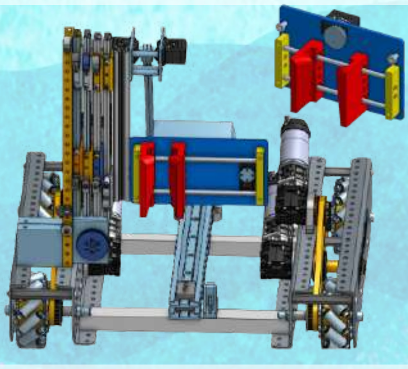
1 Dual Slider with Combined Functionality

We initially designed a dual-slider system positioned in parallel, with an intake mechanism mounted at the end. This setup was intended to allow both intake and scoring within a single subsystem, simplifying the robot's structure while enhancing overall efficiency and functionality.

2 Strengthening with Separate Subsystems

In this version, we separated the intake and scoring systems and strengthened the scoring mechanism with dual sliders. While this increased durability, it required an entirely new linear extension system for the intake, making the design more complex than anticipated. To simplify, we abandoned the extension and moved towards an even more efficient and focused scoring mechanism in the next stage.



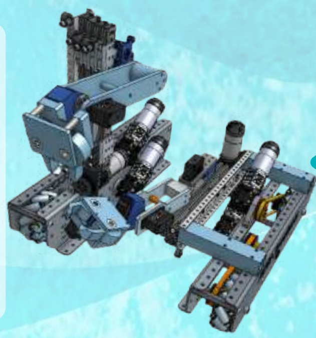


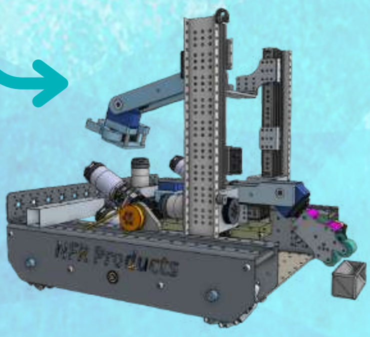
3 Passive Scoring Simplicity

This iteration introduced a mechanically simple and reliable specimen scorer, using two pieces that opened during intake and closed automatically with rubber bands. This removed the need for motorized control, reducing error and increasing scoring speed. While this solved scoring stability, we still needed better autonomous alignment and driver usability, pushing us toward the next major refinement.

4 Roller-Based Active Intake

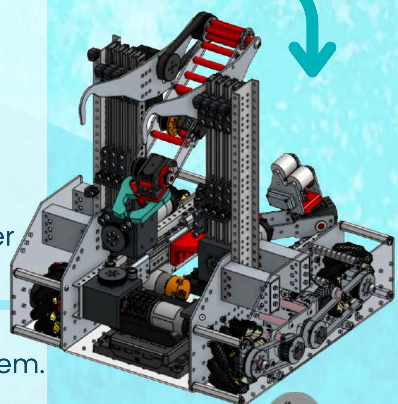
After feedback from regional competitions, we upgraded the Intake system with a roller-based design for touch-and-grab functionality and improved our autonomous to achieve a more consistent 4-piece run. We also added a parallel dual-slider system to the In-Drop for better rigidity, and integrated bearings and a new limit switch to prevent mechanical drift. These changes dramatically improved field performance and system responsiveness.





5 Lightweight Redesign & Precision Upgrades

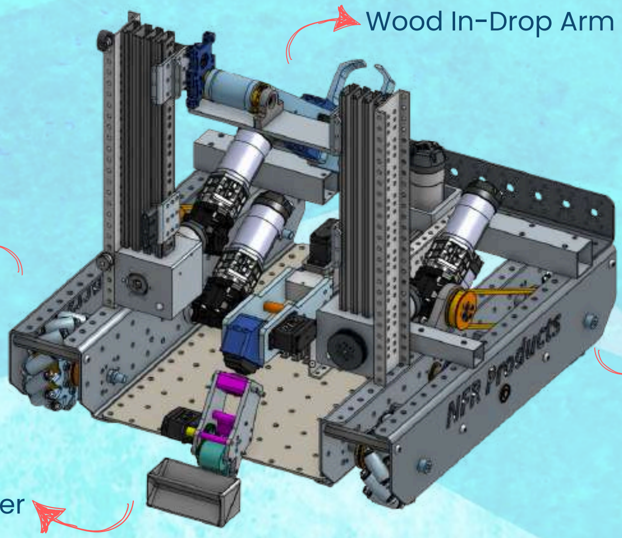
In our final iteration, we recognized that the previous chassis was too heavy, limiting our ability to perform high hangs. To solve this, we CNC-manufactured a custom lightweight chassis, increasing our robot's agility and allowing for successful endgame climbs. Alongside this, we enhanced both the Intake and In-Drop mechanisms for smoother transitions and faster scoring.



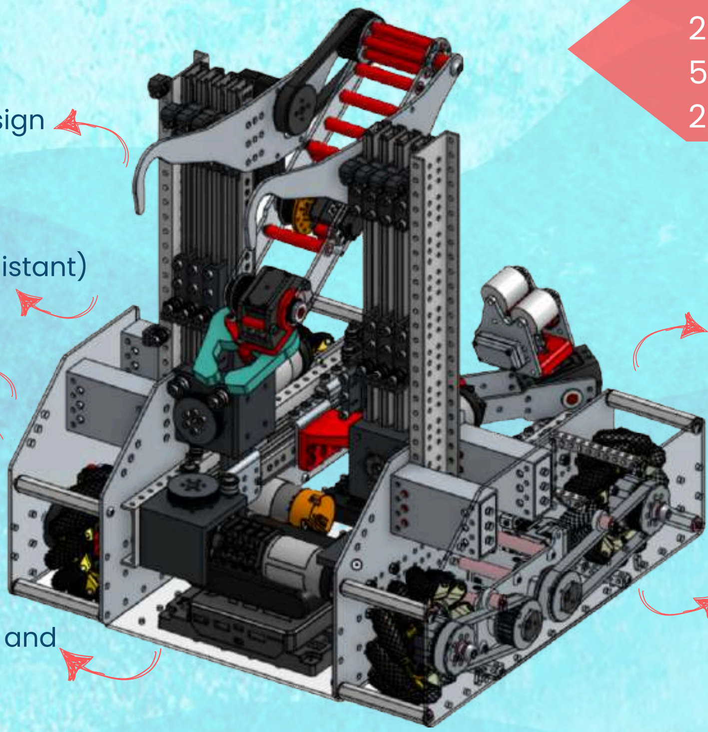
To improve positional accuracy, we replaced our optical odometry, which was highly dependent on field surface, with a more reliable dead wheel tracking system. We also added a passive hanging mechanism that enabled fast and consistent high hangs.

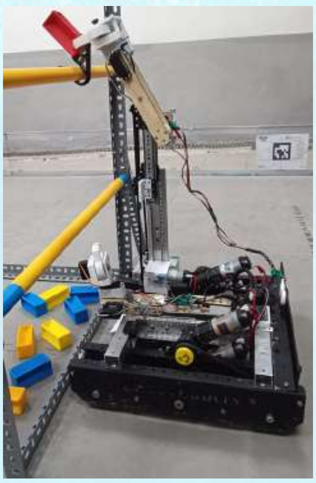
HİLDA V.II

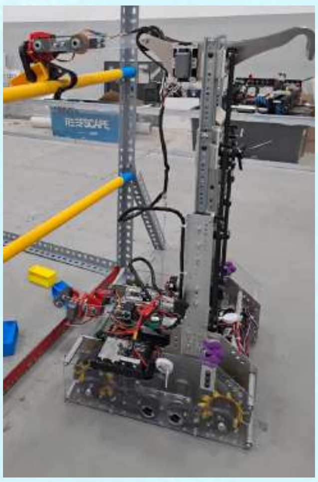
We named our robot Hilda in honor of one of the Mediterranean monk seals we adopted. Hilda is not just a robot; she is a symbol of our commitment to creating meaningful impact both on and beyond the field.



3+ Km Filament
200+ Hours 3D Design
500+ Hours Assembling
2000+ Line Code







Hilda is the newest evolution of our robot, built upon everything we learned during the Turkey Championship. While the core functions remain the same—collecting and hanging specimens—the way she performs them has been significantly improved. Unlike our previous robot, Hilda features a lighter, more compact chassis that enables a high hang, giving us a crucial scoring advantage in the endgame. We also redesigned the specimen scoring method: instead of performing an up-and-down slider motion, Hilda now scores by advancing directly forward, making the action faster, smoother, and easier to control. Every detail has been carefully optimized—not just to function, but to perform with precision and purpose.

SUBSYSTEMS

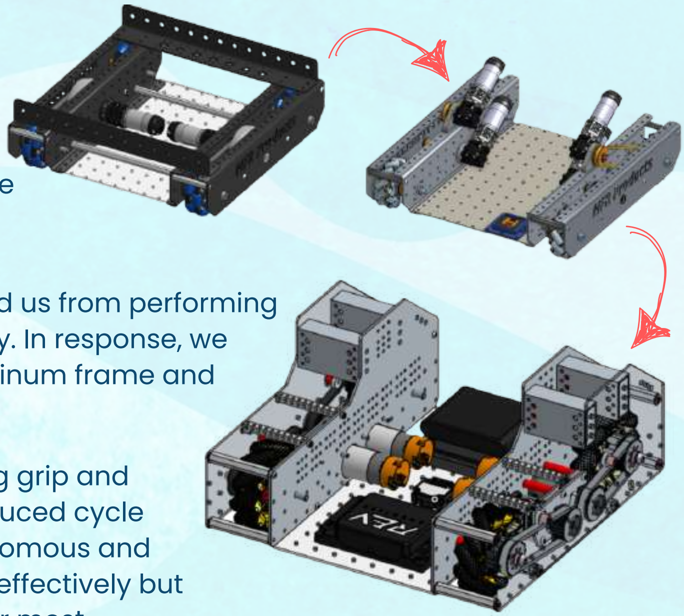
Behind every smooth motion and successful score lies a network of carefully designed and fine-tuned subsystems. At Cartesian Robotics, we treat each subsystem not as an isolated part, but as a piece of a unified machine—each one optimized for speed, stability, and synergy. From intake to hanging, every mechanism on our robot has gone through multiple iterations to ensure reliability under pressure and adaptability during matches. This section breaks down the mechanical and electronic systems that power our robot's performance and make Hilda more than just a sum of her parts.

Chasis #1

We began this season by transitioning from last year's omniwheel tank drive to a mecanum-wheeled chassis, aiming for greater maneuverability and better subsystem integration. Motors were reoriented to open up space inside the robot, improving layout efficiency and supporting our dual-slider intake design.

However, our initial chassis was too heavy, which prevented us from performing a successful hang—a major limitation in endgame strategy. In response, we developed a custom lightweight chassis with a 3mm aluminum frame and polycarbonate shell, reducing both size and weight.

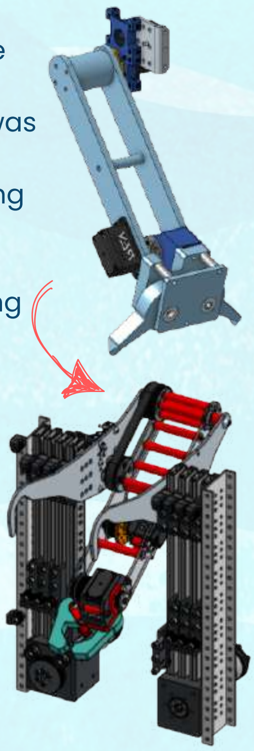
We also upgraded to GoBilda mecanum wheels, increasing grip and stability on the field. This enhanced driving control and reduced cycle times, giving our drivers better handling during both autonomous and TeleOp. The final chassis not only supports all subsystems effectively but also enabled a fast, reliable high hang, making it one of our most strategic improvements this season.



Latest Chasis
Produced by Cartesian Robotics

In-Drop #2

The In-Drop subsystem went through multiple iterations to become faster, more stable, and reliable under pressure. In earlier versions, it was mounted on a single slider, which caused noticeable wobble and made precision scoring difficult. To fix this, we transitioned to a dual parallel 4-stage slider system, which greatly improved rigidity and control, especially during high-speed cycles.

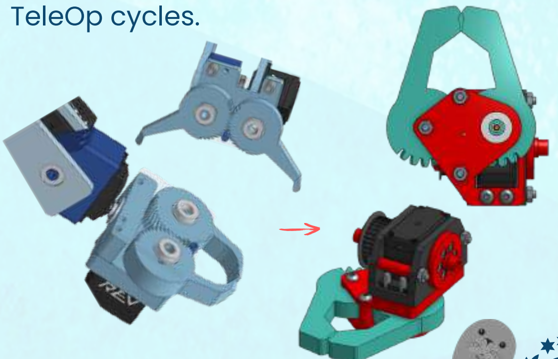


We also changed the main material of the In-Drop from wood to polycarbonate, increasing its durability and resistance to impact. This allowed us to maintain accuracy even during intense matches, while also reducing structural failures that we faced in earlier designs. The refined mechanism is now optimized to score specimens efficiently in the high chamber using a forward motion, eliminating the need for vertical movements and reducing scoring time.

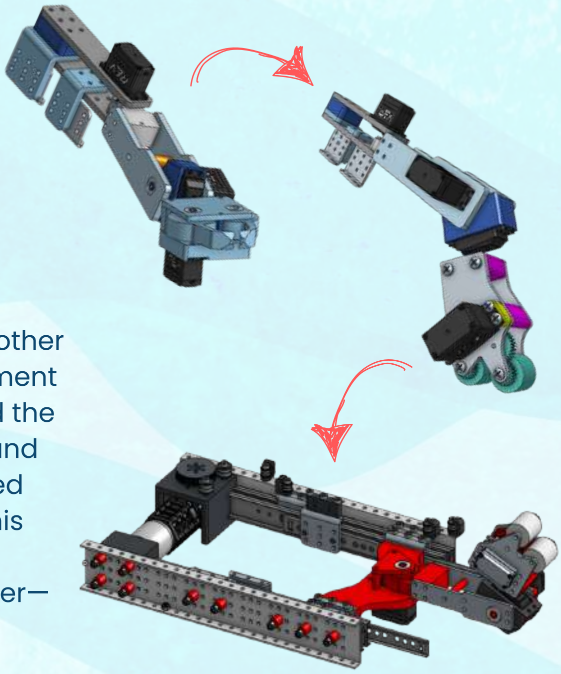
In-Drop Grabber

Our original In-Drop grabber struggled with grip reliability and alignment, especially when the intake handed over specimens at awkward angles. To address this, we designed a servo-powered grabber using a herringbone gear structure, allowing for smoother opening and closing, as well as stronger and more consistent clamping force.

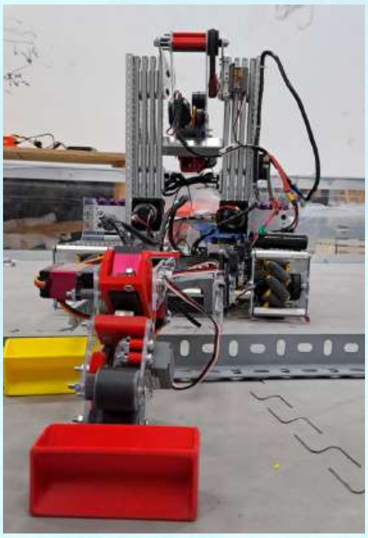
This mechanism was fine-tuned through multiple iterations to improve specimen centering, transfer reliability, and overall coordination with the intake. The result is a compact, servo-efficient grabber that enables fast pickup and seamless scoring—especially critical when executing our 5-piece autonomous and TeleOp cycles.



Intake #4



Our intake system began with a single 4-stage linear slider, which, while offering long extension, introduced significant wobble during operation. This instability made precise handoffs and fast driving difficult, especially in high-pressure matches. To solve this, we transitioned to a dual 3-stage slider design, prioritizing rigidity, speed, and driver control over maximum reach.

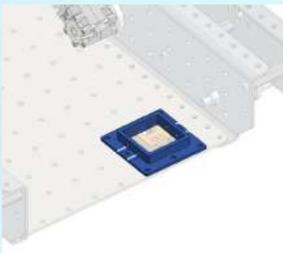


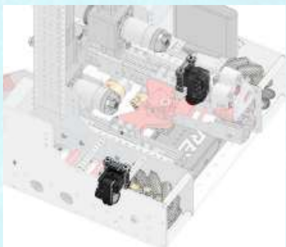
The redesign significantly improved stability and accuracy, enabling smoother specimen collection and faster alignment with the In-Drop. We also restructured the mounting to make it more compact and reliable under repeated use. Combined with our optimized intake grabbers, this system increased the number of specimens we could collect and deliver—especially noticeable in our improved autonomous consistency.

This final intake version supports faster field cycles, better alignment, and reduced driver correction, making it a key contributor to our robot's overall match efficiency.

Odometry #5

We initially used an optical odometry system, but its performance was highly dependent on field surface texture and robot height, which led to inconsistent tracking and increased error rates. To resolve these issues, we transitioned to a dead wheel tracking system, which provided more reliable and precise localization. This change significantly improved our autonomous pathing accuracy and reduced positional drift throughout the match.

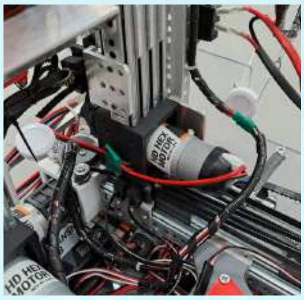




Cable Management : Card Yoyo #6

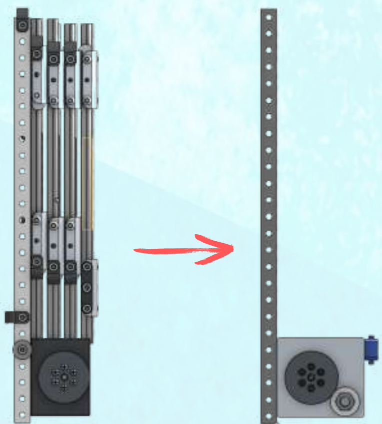
To optimize cable organization and prevent tangling during operation, we implemented a badge reel (cardholder yoyo) system. This simple yet highly effective solution kept our wires secure, retractable, and out of the way, enabling smoother subsystem movement and minimizing mechanical interference.

By integrating this method, we significantly improved the robot's reliability, safety, and long-term durability.



Sliders #7

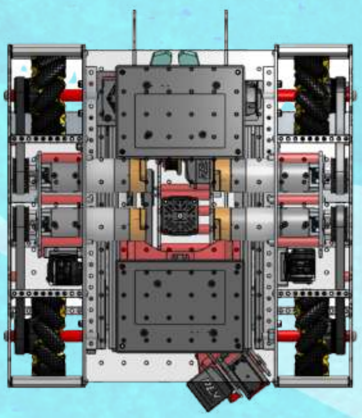
This year, we used locally manufactured sliders for the first time, which were still in the testing phase. From day one, we worked closely with the manufacturer to address performance issues and refine the design. With our input and permission, the final version included a bearing-equipped motor holder and a custom-designed limit switch, both developed by our team. The bearings eliminated gear slippage, while the limit switch ensured consistent position resets, improving overall reliability. We also took pride in sharing these innovations with other teams, contributing to collective growth within the community.



Drivetrain

Our drivetrain utilizes a mecanum wheel chassis, allowing omnidirectional movement. Field-centric control ensures that “forward” always corresponds with the field’s orientation, regardless of the robot’s heading.

At Cartesian Robotics, we design our control systems with the goal of maximizing precision, responsiveness, and adaptability, all while simplifying the driver experience. With carefully tuned subsystems, intelligent automation, and real-time feedback mechanisms, we empower our robot to perform consistently under pressure.



This is achieved through real-time readings from the robot’s IMU, combined with joystick input and custom mathematical processing. The IMU can be reset mid-match with a single button, enabling the driver to redefine orientation during dynamic gameplay. Additionally, drivetrain behavior adjusts automatically depending on the active subsystems, reducing collision risks and increasing operational safety.

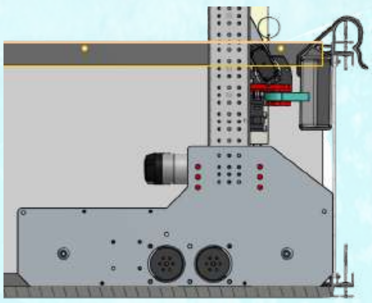
Odometry

We use a dead wheel odometry system to achieve precise localization of our robot, significantly improving consistency across different field surfaces. This system provides accuracy comparable to optical sensors while being less affected by lighting or surface texture, which is critical for both the autonomous period and automated actions during teleop. The dead wheels—unpowered omni wheels equipped with encoders—track the robot’s X and Y positions as well as rotational heading in real time. These encoder values are processed by three separate PID controllers: one each for horizontal, vertical, and rotational movement. Each controller ensures that the robot navigates smoothly and accurately to its target positions, enabling consistent and dependable performance throughout the match.

In-Drop

The In-Drop subsystem is powered by three servos and one motor, combining precision and strength for a wide range of motion. Servos are preferred in our design for their simplicity and low error rates, while the motor provides the additional force needed for tasks beyond the servo’s capacity. This motor is managed by a PID closed-loop control system, which uses encoder feedback to calculate error and adjust power output for accurate positioning.

To ensure consistent performance, we developed a custom tuning environment that allows us to test and refine PID values before implementation, minimizing the risk of mechanical issues. We also streamlined the programming process by using an enumeration-based control structure, assigning predefined functional states to the subsystem. This not only simplifies the codebase but also enhances system stability during real-time operation.



Intake

Our Intake subsystem mirrors the In-Drop system’s control structure, using four servos and one motor mounted on a horizontal slider. The similarity between the subsystems makes programming and tuning more efficient, while improving the synchronization between intake and scoring. Its three-axis arm gives the Intake system access to nearly every part of the Submersible Zone, enabling effective collection and transfer of game pieces during dynamic match conditions.

Autonomous Mode

Our autonomous system runs on a modular command structure, capable of adapting to partner strategies and match conditions. Powered by dead wheel odometry, the robot performs with high precision, enabling reliable autonomous runs. This system allows us to switch between routines, such as a 5-piece high chamber or basket strategy, without sacrificing timing or coordination.

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Teleoperated Mode

In TeleOp, we focus on reducing driver workload while enhancing responsiveness. Combined commands allow the driver to perform multiple actions—like grabbing, transferring, and scoring—through single inputs. This structure improves performance consistency, especially under time pressure. By minimizing manual complexity, the system lowers the chance of human error and ensures a smoother driving experience.

Gamepad I

Gamepad II

At Cartesian Robotics, every line of code, every bolt, and every idea is built with purpose—to grow, to inspire, and to leave a mark both on the field and beyond.

ChatGPT was used during the writing and translation process, helping us craft clear, impactful content and seamlessly switch between Turkish and English.

Canva supported us in the design and layout of this portfolio, allowing us to visually organize our content in a creative and accessible way.